

Title

Digital adherence technology with metabolite-based verification improves TB treatment completion in resource-limited settings

Abstract

Background: TB treatment non-adherence threatens individual outcomes and public health through emergence of drug-resistant strains. Digital adherence technologies combined with objective biomarker verification may improve outcomes. **Objective:** To evaluate whether integrated digital technology (TB-TST) plus standard care improves TB treatment completion compared to standard care alone. **Methods:** A total of 68 TB patients (mean age 38 years; 64% male) in Argentina's public TB program were randomized to standard directly observed therapy (n=34) or standard care plus TB-TST mobile app with urine metabolite monitoring (n=34). Primary endpoint was TB treatment completion at 6 months. Secondary endpoints included metabolite-confirmed adherence rates, app engagement metrics, and patient satisfaction. **Results:** Treatment completion occurred in 76% of TB-TST participants versus 56% of standard care (relative risk 1.36; 95% CI 1.01-1.82; $P = 0.039$). Metabolite testing confirmed adherence on 87% of scheduled assessments in the intervention arm. App utilization was high (mean 68% of prescribed daily reminders accessed). Patients reported improved confidence in their adherence ($P = 0.008$) and strengthened therapeutic relationships with treatment supporters. No serious adverse events occurred. **Conclusion:** Integrated digital adherence technology with metabolite verification enhances TB treatment completion in pragmatic clinical settings, supporting scale-up for TB program implementation.